

HOUSING PRICE PREDICTIONS



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Problem Statement

Housing price prediction is a complex regression problem that involves estimating property values based on multiple interacting factors such as geographic location, economic conditions, and housing characteristics. Unlike simple linear relationships, real-world housing markets exhibit strong nonlinear patterns, making prediction difficult for traditional statistical methods.

Conventional approaches like linear regression assume a fixed relationship between variables, which often fails in real datasets. In housing data, factors such as median income, population density, and location interact in complex ways. For example:

- Median income affects house prices differently across regions
- Geographic location influences demand and property value
- Structural features like rooms and bedrooms do not independently determine price

These overlapping relationships create hidden patterns that traditional models cannot effectively capture.

Machine learning provides a better solution because it can learn directly from data without strict assumptions. It is capable of handling nonlinear relationships and feature interactions automatically.

- Captures complex patterns in data
- Learns interactions between multiple variables
- Improves accuracy through data-driven learning
- Works well with large and diverse datasets

This makes machine learning highly suitable for housing price prediction, where many factors jointly influence the final price.

Advanced ensemble methods such as Random Forest, Gradient Boosting, and XGBoost further improve prediction accuracy. These models combine multiple learners to reduce errors and enhance generalization. For example, Random Forest reduces overfitting, Gradient Boosting learns from previous mistakes, and XGBoost optimizes performance using efficient boosting techniques.

Housing price prediction requires advanced machine learning techniques due to the complexity and nonlinear nature of real estate data. Ensemble models provide more accurate, reliable, and interpretable results compared to traditional statistical methods, making them essential for modern predictive systems in real estate analysis.

Objective

The objectives of this project are:

1. To analyze the features of the housing dataset and understand their relationships.
2. To preprocess the data by handling missing values, encoding categorical variables, and applying feature scaling.
3. To perform exploratory data analysis (EDA) to understand data distribution and patterns.
4. To train multiple regression models for predicting housing prices.
5. To compare the performance of different models using evaluation metrics.
6. To improve model accuracy through hyperparameter tuning techniques.
7. To identify the best-performing machine learning model for final prediction.

Dataset Description

The dataset used in this project consists of **20,640 samples** and **10 features**, including both numerical and categorical variables. It represents housing data that captures different geographic, economic, and structural characteristics of properties, which are used to predict housing prices.

Features:

1. **longitude**: Represents the geographical east-west position of a house.
2. **latitude**: Represents the geographical north-south position of a house.
3. **housing_median_age**: Indicates the median age of houses in a given area.
4. **total_rooms**: Total number of rooms available in a housing block.
5. **total_bedrooms**: Total number of bedrooms in a housing block.
6. **population**: Total population living in the area.
7. **households**: Number of households in the region.
8. **median_income**: Median income of households in the area, an important economic factor.
9. **ocean_proximity**: A categorical feature representing the location of houses relative to the ocean (e.g., near ocean, inland, etc.).

Target Variable:

10. **median_house_value**: This is the continuous numeric variable that represents the median value of houses. It is the output variable that the machine learning models aim to predict.

Data Preprocessing

Data preprocessing was performed to clean and prepare the dataset for machine learning models. This step ensures that the data is consistent, complete, and suitable for training.

Missing Value Handling

1. The dataset contains missing values in the **total_bedrooms** feature.
2. These missing values were handled using **median imputation**, where each missing value was replaced with the median of the column.
3. This approach helps maintain data stability and reduces the impact of outliers.

Categorical Encoding

1. The categorical feature **ocean_proximity** was converted into numerical form using **One-Hot Encoding**.
2. This process creates separate binary columns for each category.
3. The **drop-first method** was applied to avoid multicollinearity and reduce redundancy in the dataset.

Feature Scaling

1. Feature scaling was applied using **StandardScaler** to normalize the range of numerical variables.
2. This step is important for models that are sensitive to feature magnitude.
3. Scaling was specifically applied for:
 - Support Vector Regression (SVR)
 - K-Nearest Neighbors (KNN)
 - Linear Regression
 - Ridge Regression

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the structure, patterns, and relationships within the dataset before model training. It helped in identifying important features and data characteristics.

Missing Value Analysis

1. Heatmaps were used to visually inspect missing values in the dataset.
2. The analysis confirmed that only the **total_bedrooms** feature contained missing values, while all other features were complete.
3. This step helped in deciding the appropriate imputation strategy.

Distribution Analysis

1. The distribution of **median_house_value** was analyzed using visual plots.
2. The data showed a **right-skewed distribution**, indicating that most houses fall within the lower to mid-price range.
3. Only a small number of houses have very high values, creating a long tail in the distribution.

Correlation Analysis

A correlation study was conducted to understand relationships between features and the target variable.

Key findings include:

1. **median_income** shows a strong positive correlation with house prices (~ 0.68), making it the most influential feature.
2. **latitude** shows a weak negative correlation (~ -0.14), indicating slight regional influence.
3. Other structural features such as rooms, bedrooms, and population show weak correlations with the target variable.

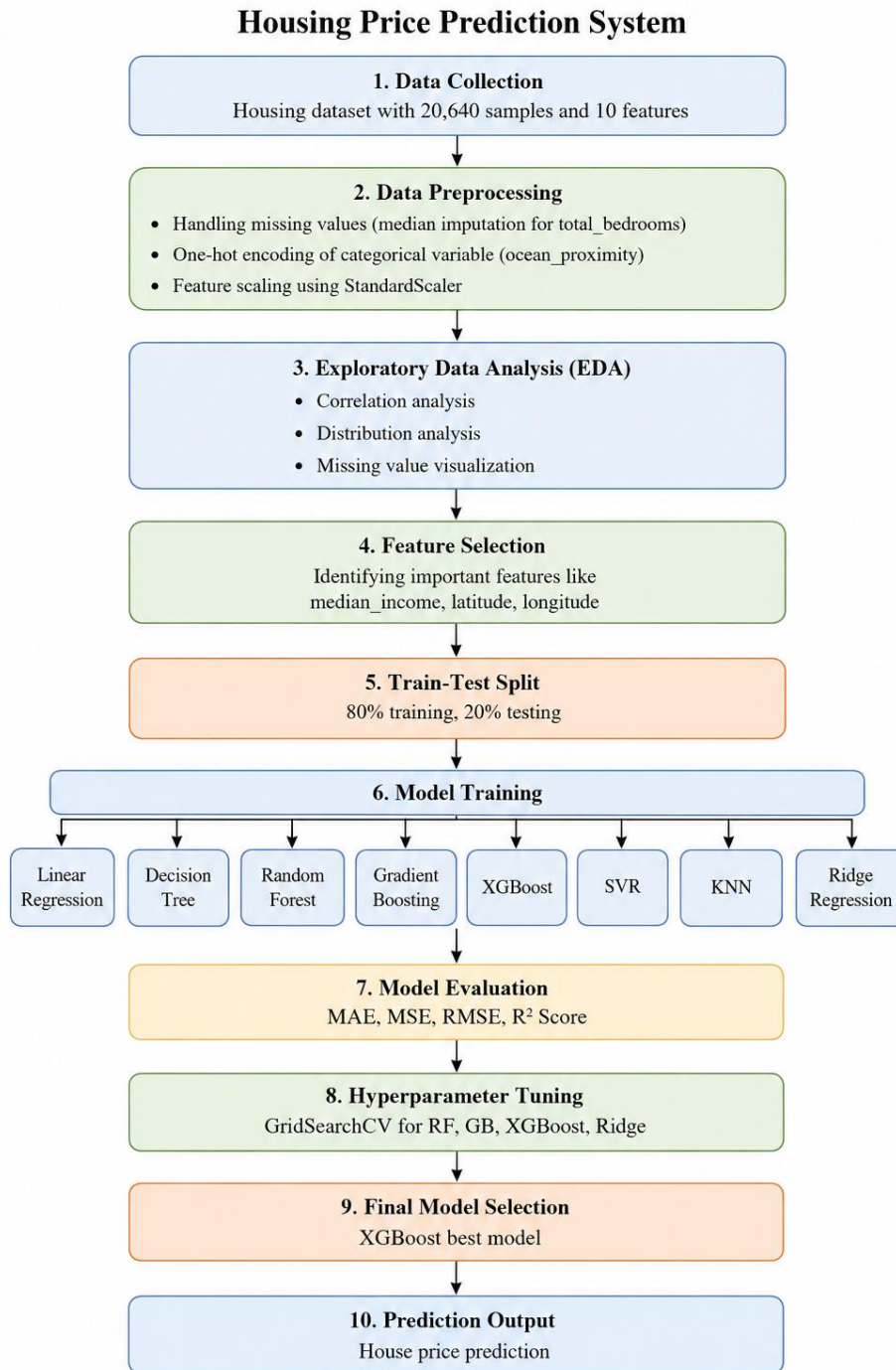
Feature Insights

Based on EDA, the most influential features affecting house prices are:

1. **median_income** (strongest predictor)
2. **longitude** (geographical impact)
3. **latitude** (regional variation)

These insights guided the feature selection and model training process.

Flow Chart



This image is AI-generated.

Feature Engineering

Feature engineering was performed to convert raw data into a format suitable for machine learning models and to improve model performance.

1. The categorical variable **ocean_proximity** was transformed using **One-Hot Encoding**, converting it into multiple binary features.
2. After encoding, the original **ocean_proximity** column was removed to avoid redundancy in the dataset.
3. This transformation ensured that the categorical information was properly represented in numerical form for model training.
4. The final processed dataset contained a shape of **(20640, 12)**, indicating 20,640 samples and 12 features after feature expansion.

Train-Test Split

The dataset was divided into training and testing sets to evaluate model performance on unseen data.

1. **Training data:** 80% of the dataset (16,512 samples) was used to train the machine learning models.
2. **Testing data:** 20% of the dataset (4,128 samples) was reserved for evaluating model performance.
3. A fixed **random state** was used to ensure reproducibility of the results, meaning the same split can be obtained every time the code is executed.

Machine Learning Models

The following regression models were used in this project to predict housing prices and compare their performance.

Linear Regression

This is a baseline model that assumes a linear relationship between input features and the target variable. It is simple and easy to interpret but may not capture complex patterns in the data.

Decision Tree Regressor

This model splits the data into decision rules and can capture nonlinear relationships. However, it is prone to overfitting if not properly tuned.

Random Forest Regressor

An ensemble learning method that combines multiple decision trees to improve accuracy. It reduces overfitting and provides more stable predictions.

Gradient Boosting Regressor

A sequential learning technique where each new model corrects the errors of the previous one, leading to improved performance.

XGBoost Regressor

An advanced and optimized implementation of gradient boosting that offers high accuracy, speed, and regularization for better performance.

Support Vector Regressor (SVR)

A kernel-based regression model that works well for complex relationships but is sensitive to feature scaling.

K-Nearest Neighbors (KNN)

A distance-based algorithm that predicts values based on the closest data points in the feature space.

Model Evaluation Metrics

Models were evaluated using standard regression performance metrics to measure prediction accuracy and error magnitude.

Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values. It shows how far predictions are from actual values on average.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Mean Squared Error (MSE)

MSE calculates the average of squared differences between actual and predicted values. It penalizes larger errors more heavily.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Root Mean Squared Error (RMSE)

RMSE is the square root of MSE and represents the error in the same unit as the target variable.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

R² Score (Coefficient of Determination)

R² measures how well the model explains the variance in the target variable. A value closer to 1 indicates better performance.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Where:

1. y_i = actual value
2. $\hat{y}_i$ = predicted value
3. $\bar{y}$ = mean of actual values
4. n = number of samples

Model Performance Comparison

MODELS NAME	R ² SCORE	RMS E	MAE	MODEL FITTING	PERFOMANCE
<i>XG BOOST</i>	<i>0.829</i>	47309	31645	<i>GOOD</i>	<i>BEST</i>
<i>DESICION TREE</i>	<i>0.628</i>	49038	31639	<i>SLIGHTLY OVERFIT</i>	<i>COULD BE BETTER</i>
<i>RANDOM FOREST</i>	<i>0.816</i>	55884	38248	<i>GOOD</i>	<i>NICE</i>
<i>GRADIENT BOOSTING</i>	<i>0.761</i>	61367	40854	<i>GOOD</i>	<i>NICE</i>
<i>LINEAR REGRESSIO N</i>	<i>0.625</i>	69755	44180	<i>SLIGHTLY UNDERFIT</i>	<i>COULD BE BETTER</i>
<i>K- NEAREST NEIGHBOU R</i>	<i>0.712</i>	70060	50670	<i>GOOD</i>	<i>NICE</i>
<i>SUPPORT VECTOR REGRESSOR</i>	<i>-0.04</i>	116858	86961	<i>SEVERE UNDERFIT</i>	<i>WORST</i>

Key Observations

1. XGBoost achieved the best overall performance among all models with the highest R² score and lowest error values.
2. Support Vector Regressor (SVR) performed the worst, mainly due to its sensitivity to scaling and inability to capture complex dataset patterns effectively.

Feature Importance Analysis

The most important predictors in the model include:

1. **Median Income:** The most influential feature with the highest impact on house prices.
2. **Longitude:** Plays a significant role due to geographic location effects.
3. **Latitude:** Contributes to regional price variations across areas.
4. **Housing structural features:** Such as rooms, bedrooms, and population, have relatively lower importance in prediction.

Economic factors, especially median income, are the dominant drivers of housing price prediction compared to structural attributes of houses.

Hyper- Parameters Tuning

Random Forest

1. n_estimators: 200
2. max_depth: None
3. CV Score: ≈ 0.813

Gradient Boosting

1. n_estimators: 200
2. learning_rate: 0.1
3. max_depth: 5
4. subsample: 1.0
5. CV Score: ≈ 0.824

XGBoost

1. n_estimators: 300
2. max_depth: 8
3. learning_rate: 0.05
4. subsample: 0.8
5. CV Score: ≈ 0.833

Ridge Regression

1. alpha: 10
2. CV Score: ≈ 0.647

Model Comparison (Before vs After Tuning)

After hyperparameter tuning, the performance of different models showed noticeable changes in accuracy and generalization ability. Overall, ensemble models benefited the most from tuning, while linear models showed limited improvement due to their inherent constraints.

1. XGBoost showed a clear improvement in accuracy after tuning, achieving better predictive performance and stability.
2. Random Forest improved slightly, indicating that optimized parameters helped enhance its generalization ability.
3. Ridge Regression showed minimal improvement because it is limited by its linear assumptions and cannot capture complex nonlinear relationships in the data effectively.

These results highlight that advanced ensemble models respond better to hyperparameter optimization compared to simpler linear models.

Final Model Selection

The best-performing model is the **Tuned XGBoost Regressor**, which achieved the highest overall accuracy among all tested models.

Final performance metrics include:

1. MAE ≈ 30087
2. RMSE ≈ 46434
3. R² Score ≈ 0.835

Key reasons for selecting this model are:

1. It achieved the highest predictive accuracy compared to other models.
2. It demonstrates strong generalization on unseen data.
3. It effectively captures complex nonlinear relationships in the dataset.

Outputs, Graphs and Visualizations

The project includes the following visual analyses:

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity
0	-122.23	37.88	41.0	880.0	129.0	322.0	126.0	8.3252	452600.0	NEAR BAY
1	-122.22	37.86	21.0	7099.0	1106.0	2401.0	1138.0	8.3014	358500.0	NEAR BAY
2	-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100.0	NEAR BAY
3	-122.25	37.85	52.0	1274.0	235.0	558.0	219.0	5.6431	341300.0	NEAR BAY
4	-122.25	37.85	52.0	1627.0	280.0	565.0	259.0	3.8462	342200.0	NEAR BAY

Chart 1: Head Values of Dataset

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity
20635	-121.09	39.48	25.0	1665.0	374.0	845.0	330.0	1.5603	78100.0	INLAND
20636	-121.21	39.49	18.0	697.0	150.0	356.0	114.0	2.5568	77100.0	INLAND
20637	-121.22	39.43	17.0	2254.0	485.0	1007.0	433.0	1.7000	92300.0	INLAND
20638	-121.32	39.43	18.0	1860.0	409.0	741.0	349.0	1.8672	84700.0	INLAND
20639	-121.24	39.37	16.0	2785.0	616.0	1387.0	530.0	2.3886	89400.0	INLAND

Chart 2: Tail Values of Dataset

```
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  -
0   longitude            20640 non-null  float64
1   latitude             20640 non-null  float64
2   housing_median_age   20640 non-null  float64
3   total_rooms          20640 non-null  float64
4   total_bedrooms       20433 non-null  float64
5   population           20640 non-null  float64
6   households           20640 non-null  float64
7   median_income        20640 non-null  float64
8   median_house_value   20640 non-null  float64
9   ocean_proximity      20640 non-null  object
dtypes: float64(9), object(1)
```

Chart 3: Dataset Basic Information

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	0
longitude	0
latitude	0
housing_median_age	0
total_rooms	0
total_bedrooms	207
population	0
households	0
median_income	0
median_house_value	0
ocean_proximity	0
dtype: int64	

Chart 4: Identifying the null values

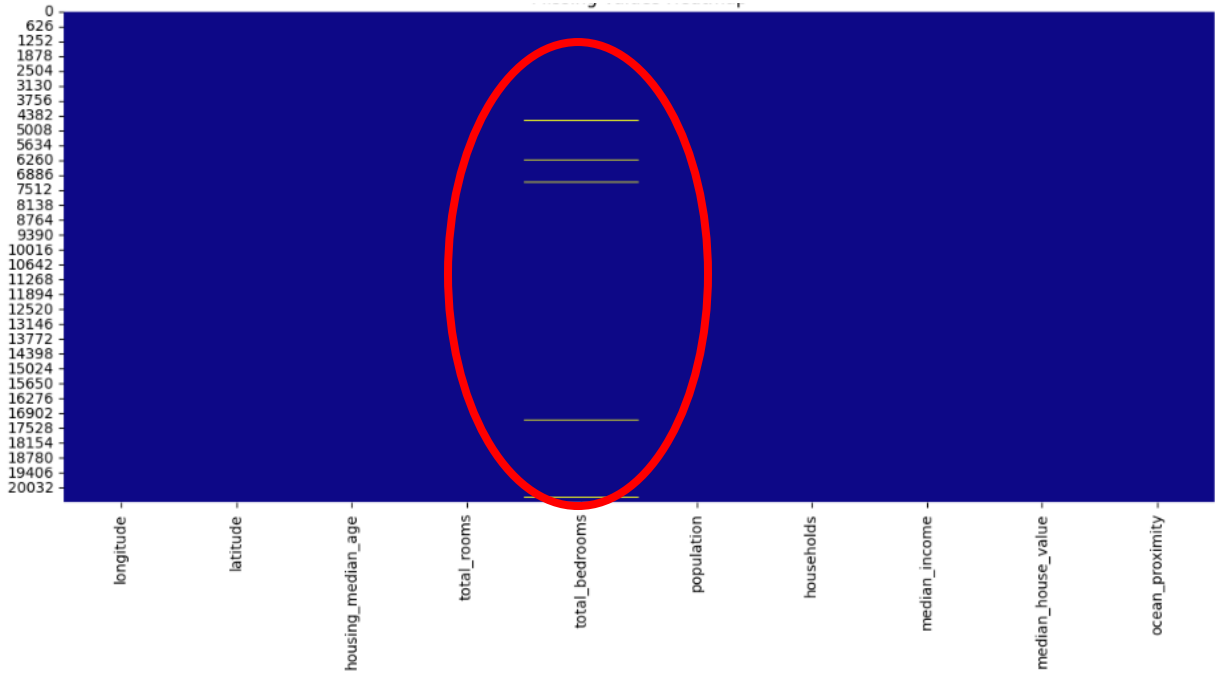


Figure 1: Missing values distribution heatmap

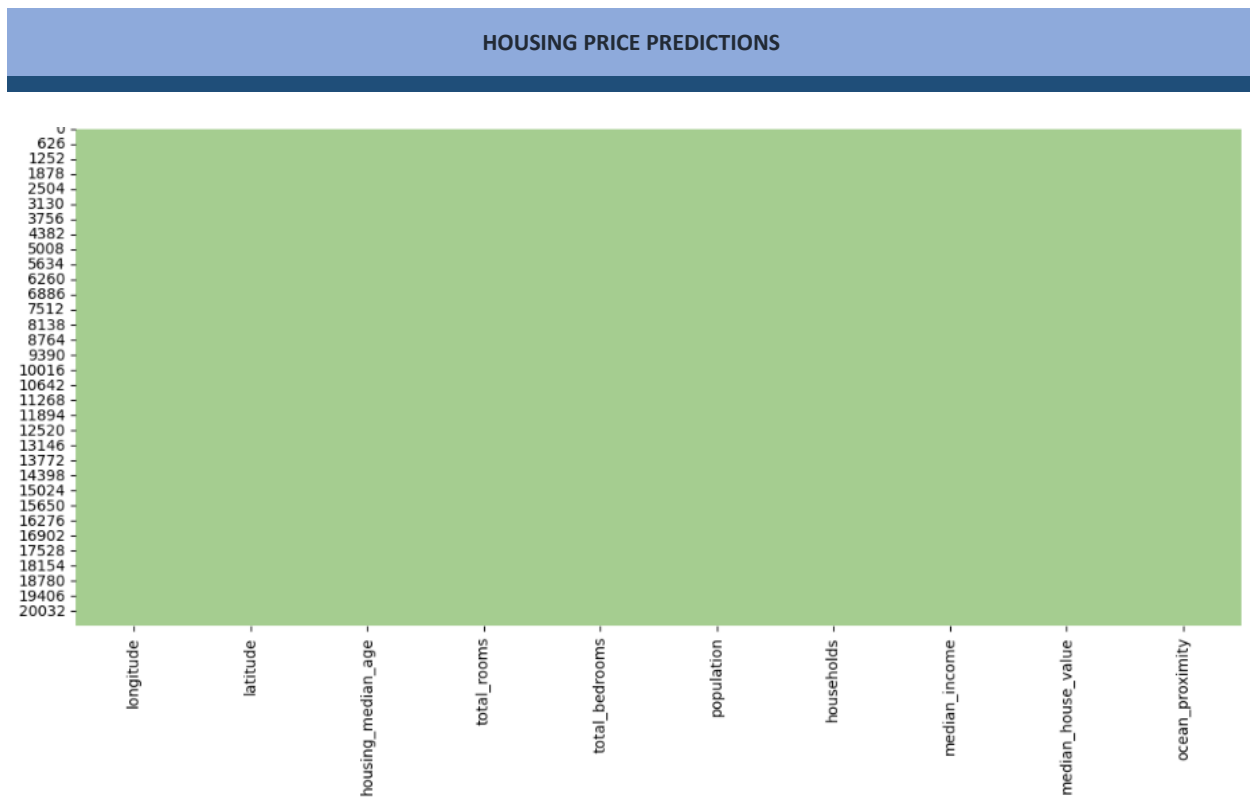


Figure 2: Heatmap Visualization After Missing Values Imputation

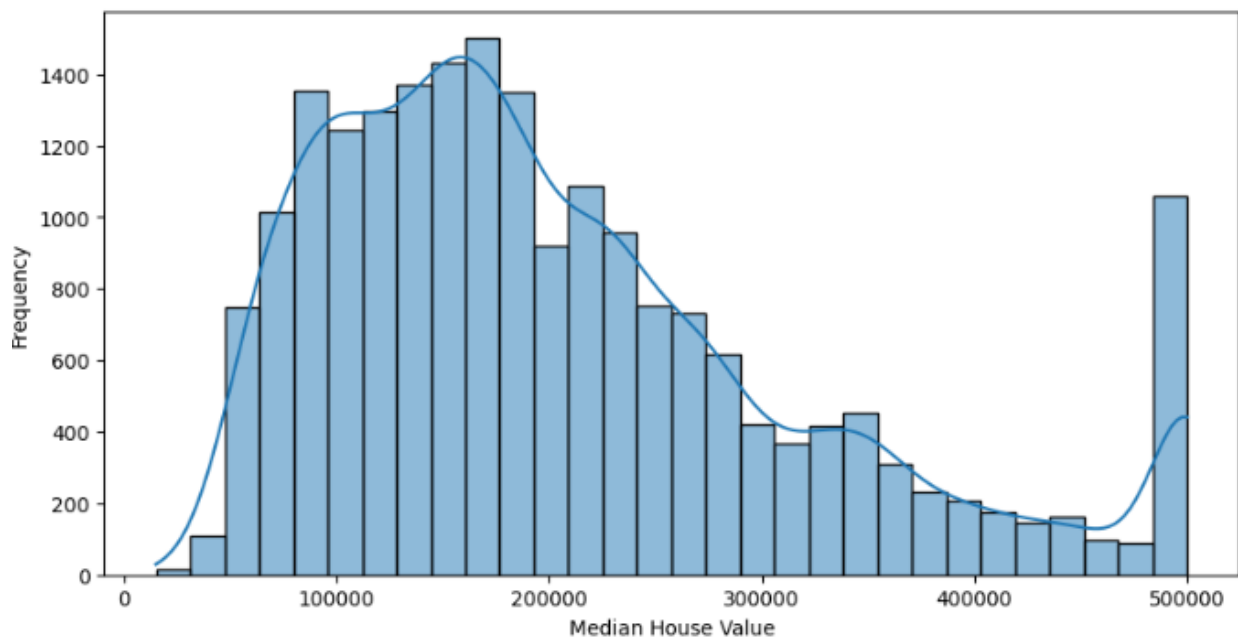


Figure 3: Distribution of Housing Prices

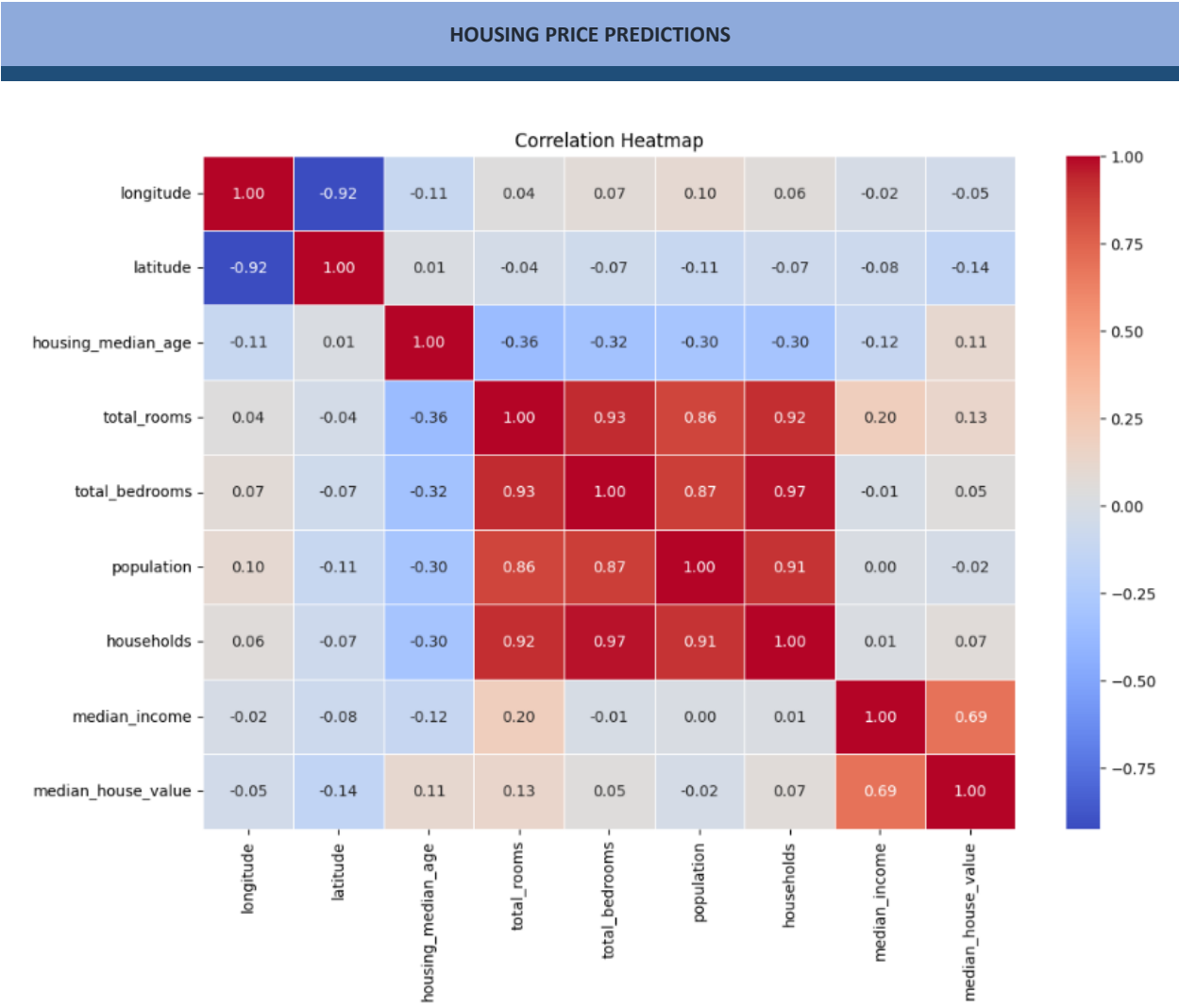


Figure 4: Correlation Heatmap

```
median_house_value    1.000000
median_income          0.688075
total_rooms            0.134153
housing_median_age     0.105623
households             0.065843
total_bedrooms         0.049457
population             -0.024650
longitude              -0.045967
latitude               -0.144160
Name: median_house_value, dtype: float64
```

Chart 5: Target Feature Selection

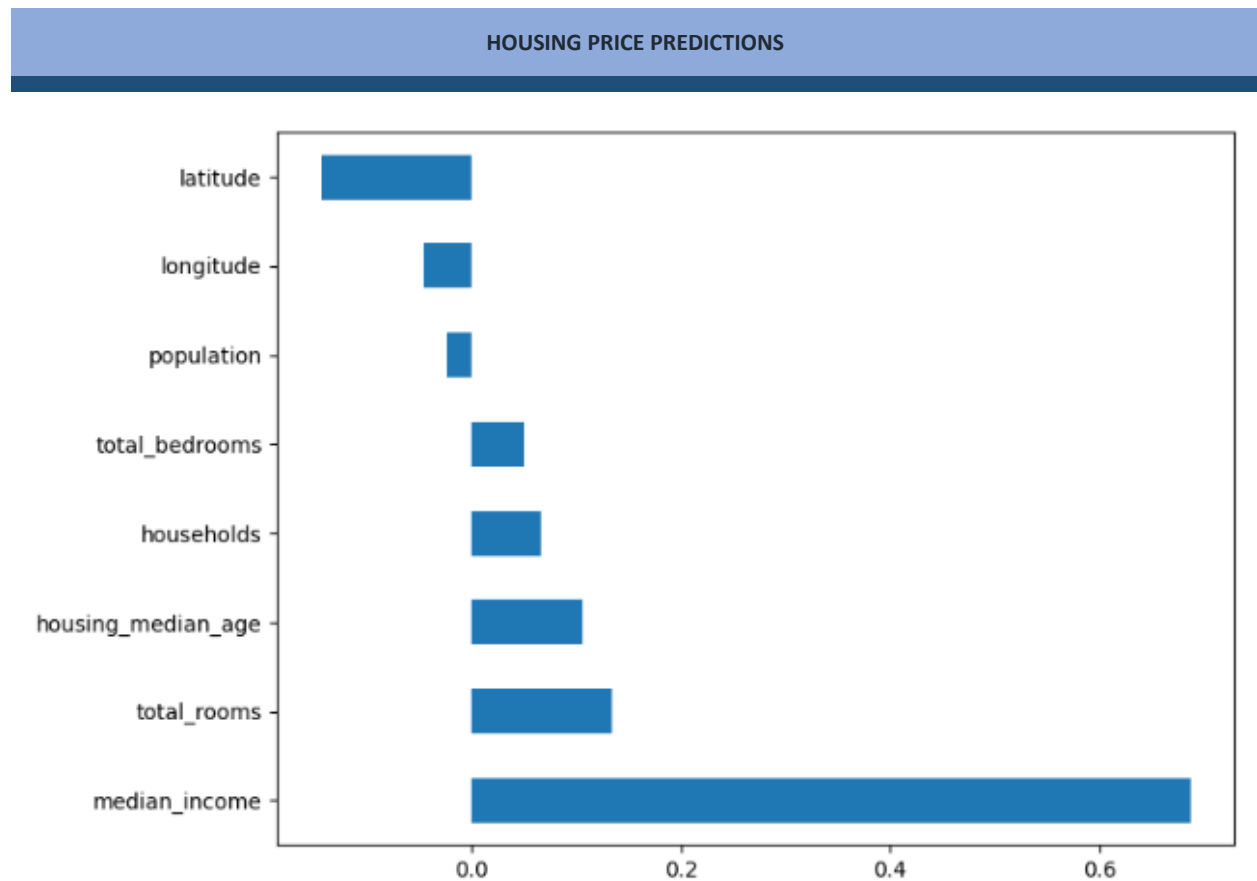


Figure 5: Feature Importance, Correlation with Target

	Model	MAE	MSE	RMSE	R2 Score
4	XGBoost	31645.725504	2.238154e+09	47309.128542	0.829202
2	Random Forest	31639.373406	2.404746e+09	49038.209338	0.816489
3	Gradient Boosting	38248.031950	3.123095e+09	55884.659003	0.761670
6	KNN	40854.397771	3.765913e+09	61367.037403	0.712615
1	Decision Tree	44180.854409	4.865869e+09	69755.779954	0.628676
0	Linear Regression	50670.738241	4.908477e+09	70060.521845	0.625424
5	SVR	86961.497845	1.365597e+10	116858.775898	-0.042115

Chart 6: Models Performance

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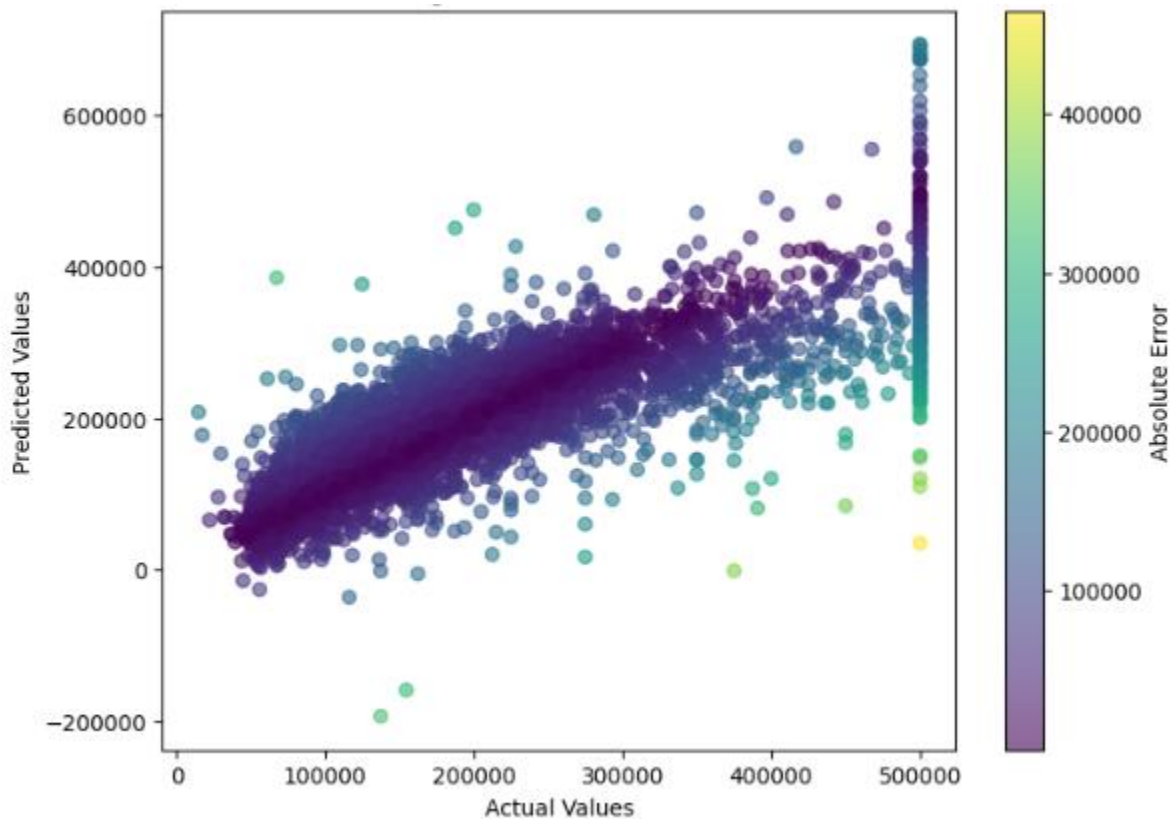


Figure 6: Linear Regression, Actual Vs Predicted Values

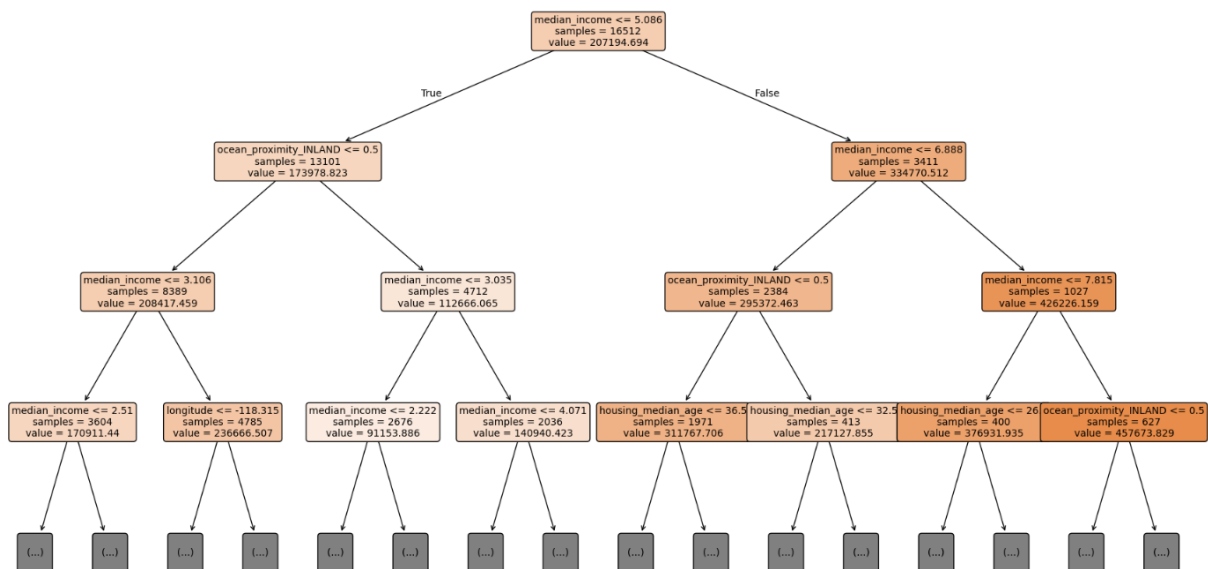


Figure 7: Decision Tree Visualization

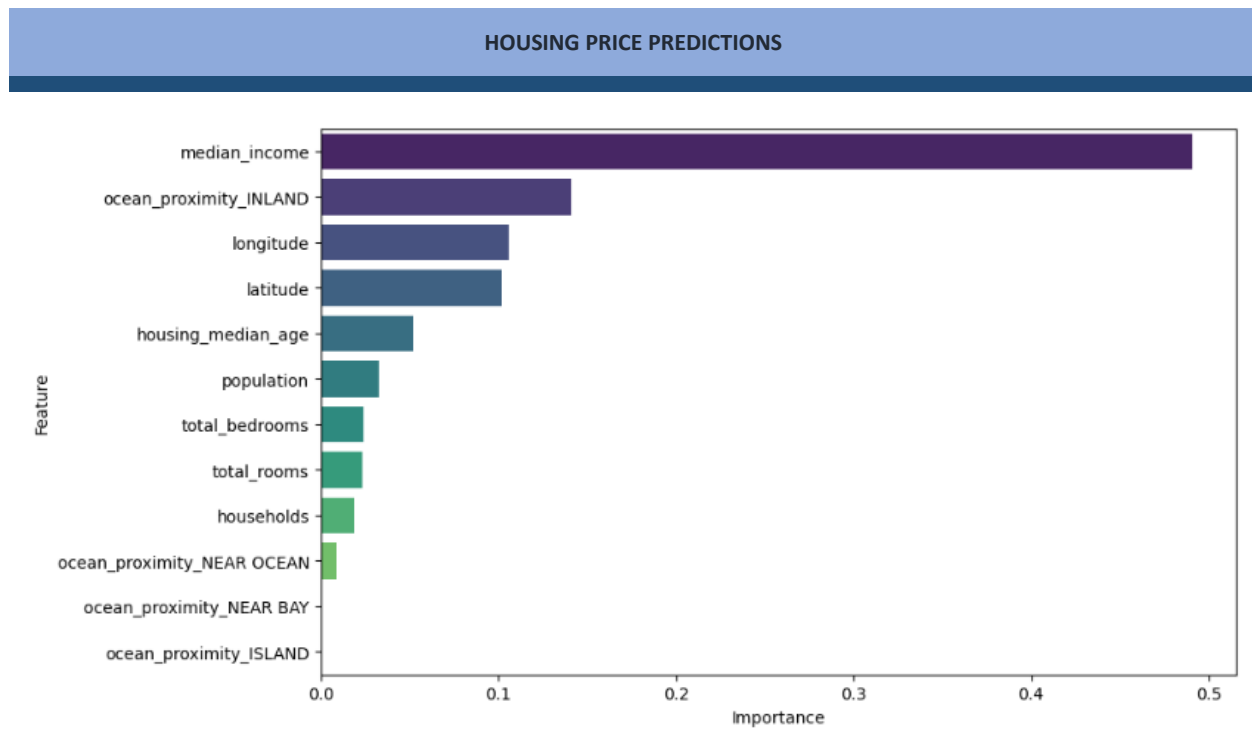


Figure 8: Random Forest Feature Importance Visual Graph

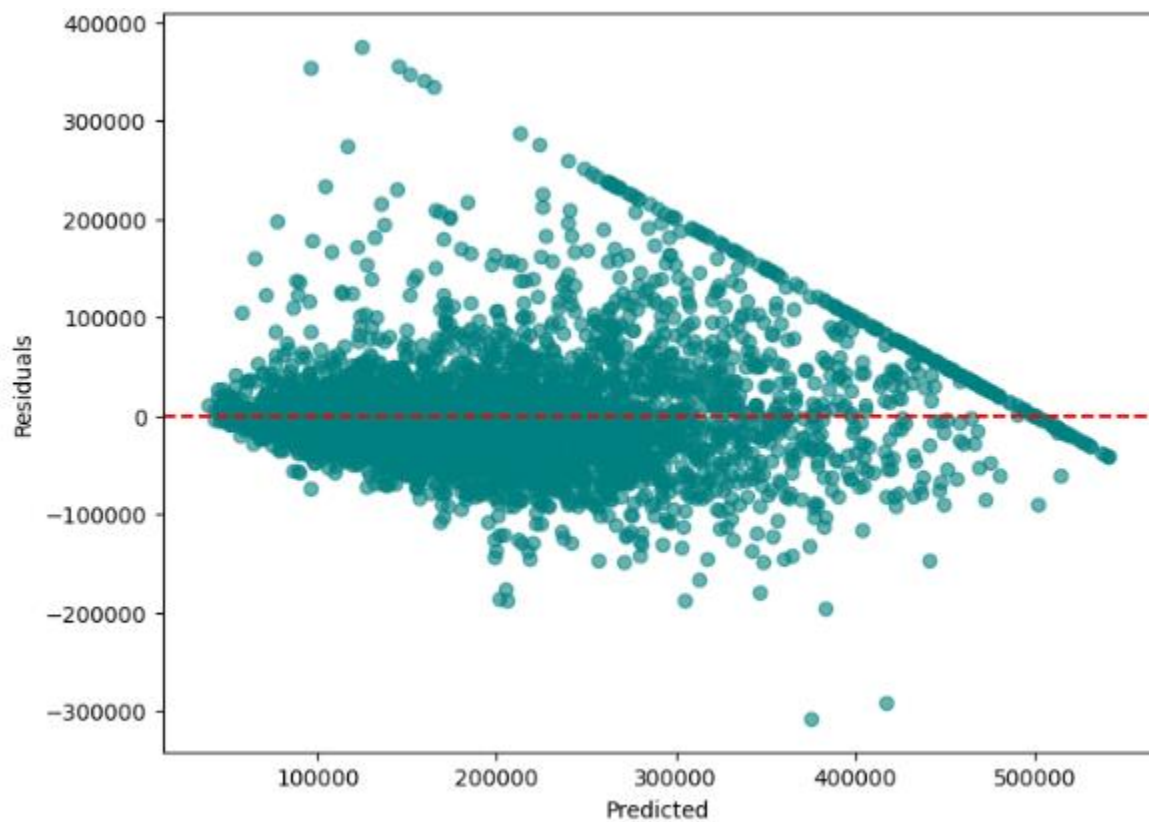


Figure 9: Gradient Boosting Residual Plot

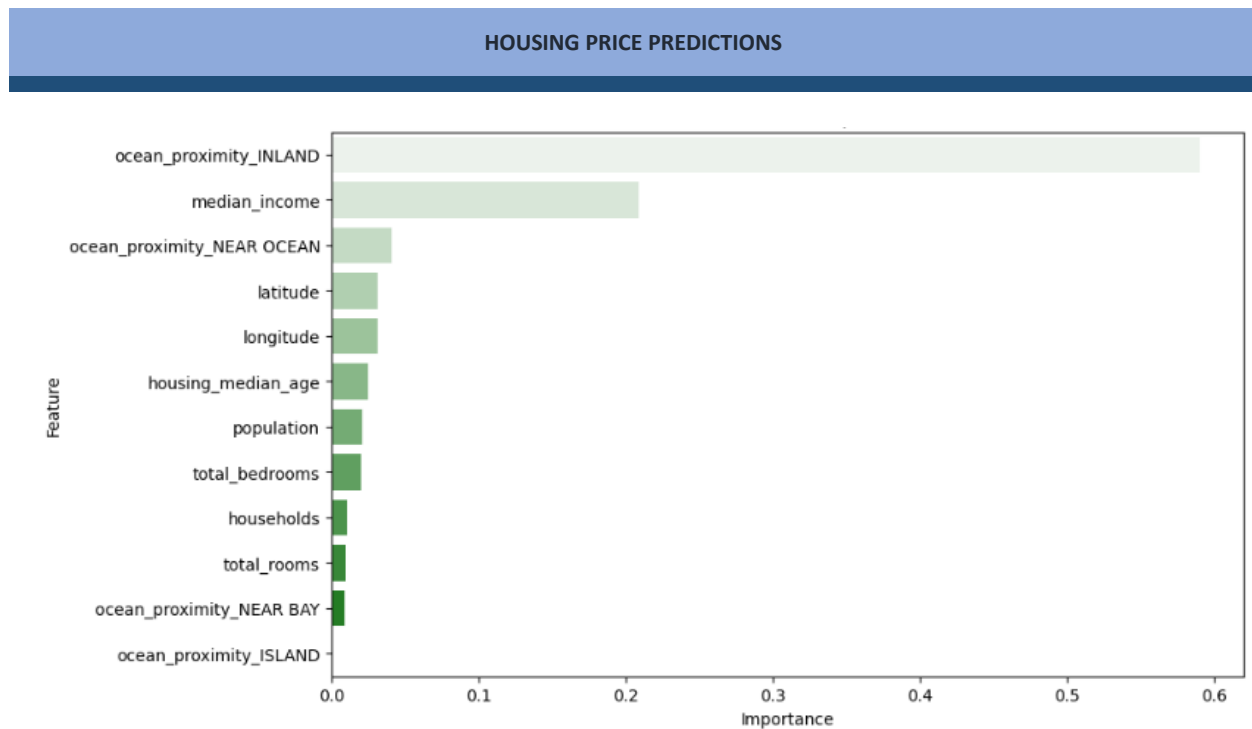


Figure 10: XG Boost Feature Importance Graph

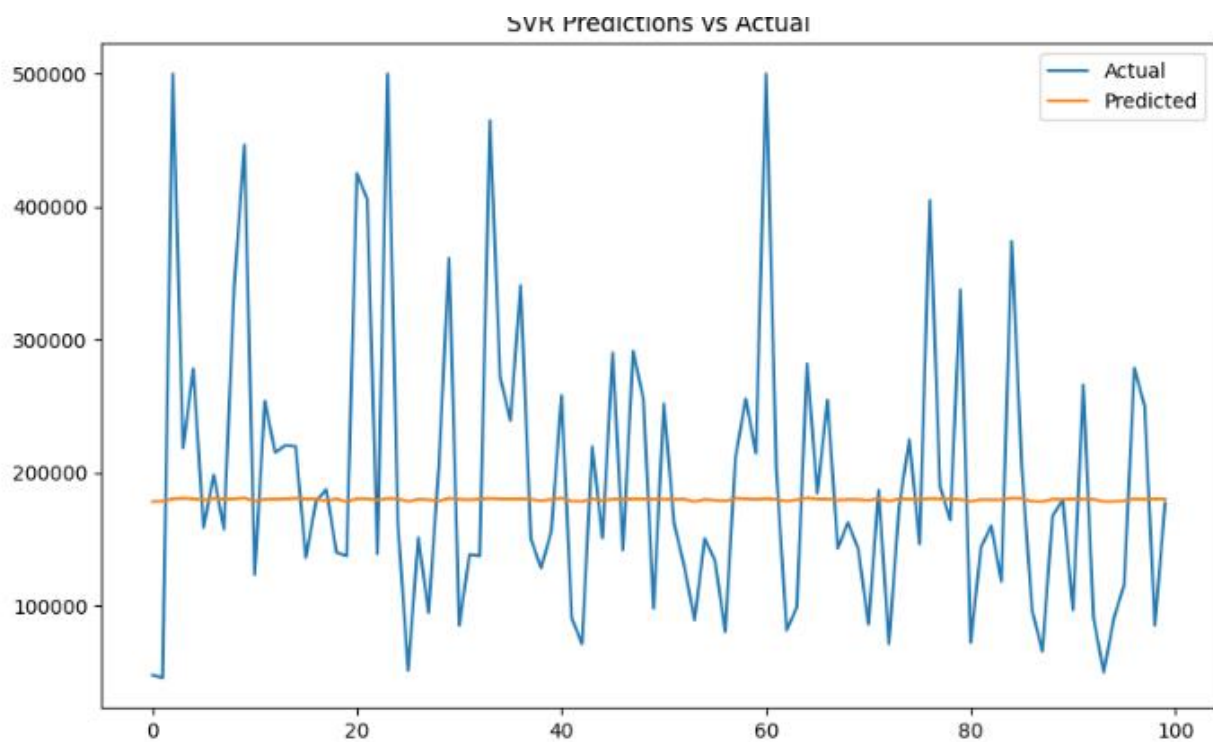


Figure 11: SVR Actual Vs Predicted Values

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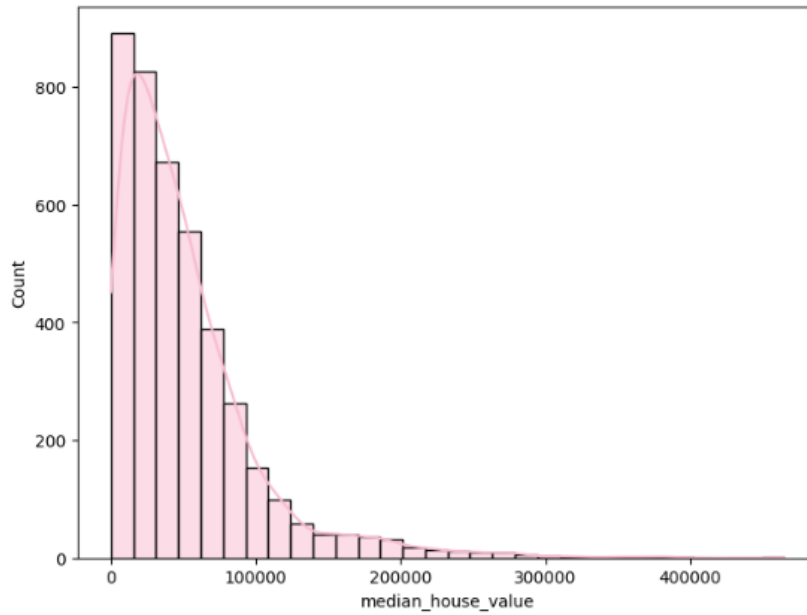


Figure 12: KNN Error Distribution Graph

	MAE (Before)	MSE (Before)	RMSE (Before)	R2 Score (Before)	Model	MAE (After)	MSE (After)	RMSE (After)	R2 Score (After)	MAE % Change	RMSE % Change	R2 % Change
0	31645.725504	2.238154e+09	47309.128542	0.829202	XGBoost	30087.044708	2.156202e+09	46434.919525	0.835456	4.925407	1.847865	0.754209
1	31639.373406	2.404746e+09	49038.209338	0.816489	Random Forest	31492.938259	2.386213e+09	48848.879242	0.817903	0.462826	0.386087	0.173216
2	38248.031950	3.123095e+09	55884.659003	0.761670	Gradient Boosting	32057.322286	2.297175e+09	47928.851646	0.824698	16.185695	14.236120	8.274930
3	50668.122931	4.908042e+09	70057.416870	0.625457	Ridge Regression	50646.170659	4.904384e+09	70031.305311	0.625736	0.043326	0.037272	0.044630

Chart 7: Tuned Models Performance

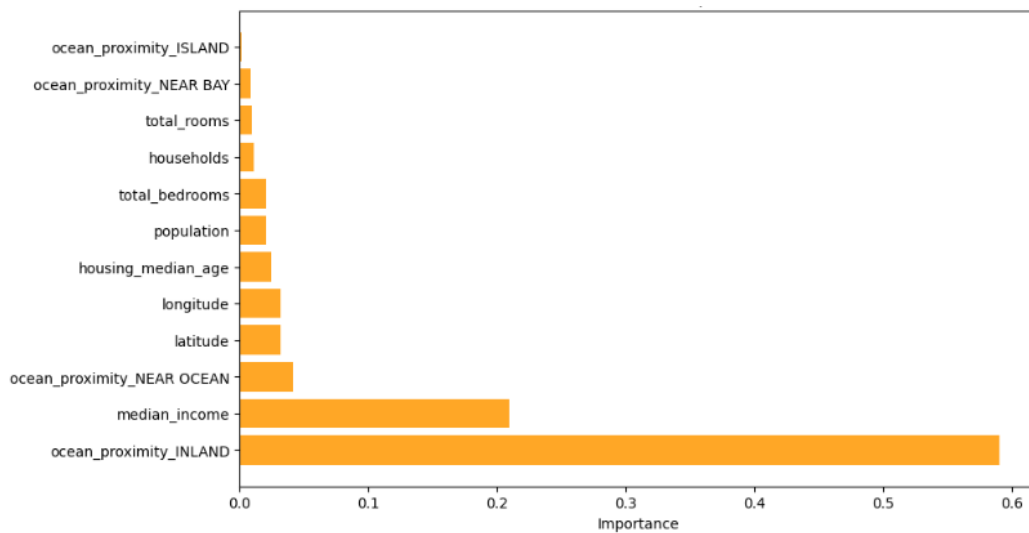


Figure 13: Tuned Random Forest Top Features

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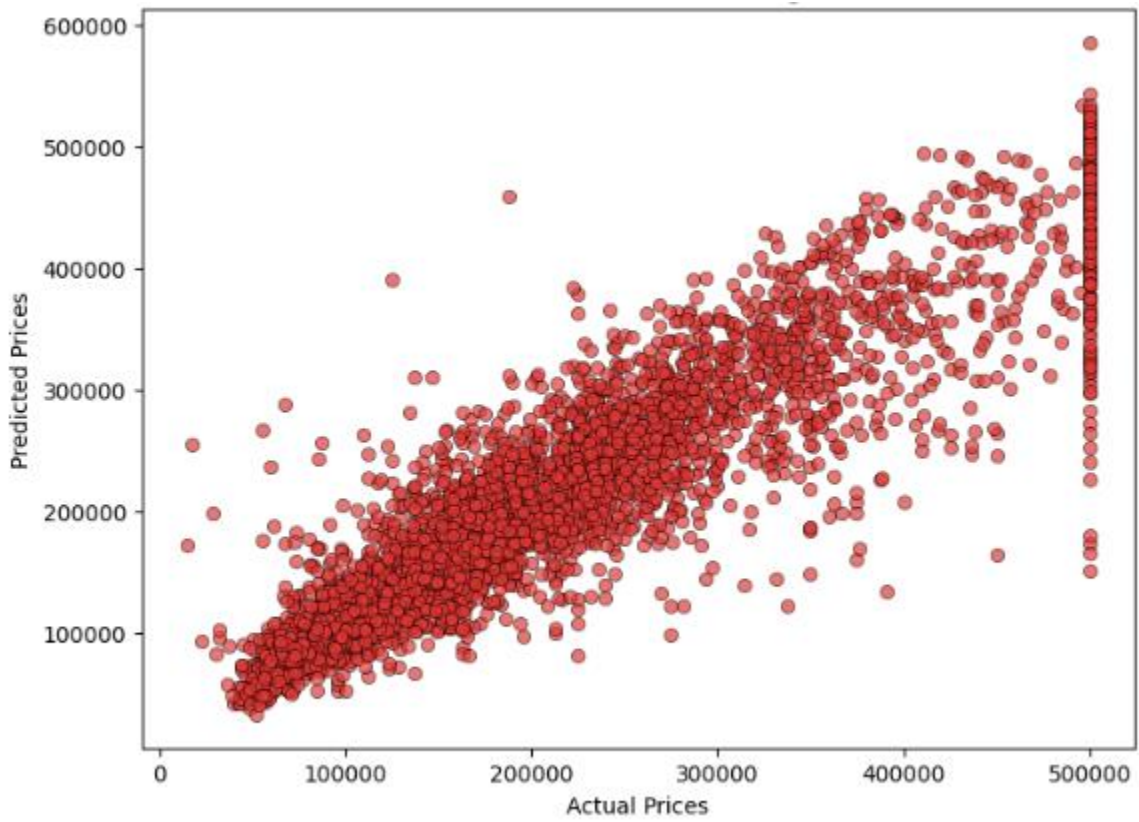


Figure 14: Tuned Gradient Boosting Actual Vs Predicted Values Graph

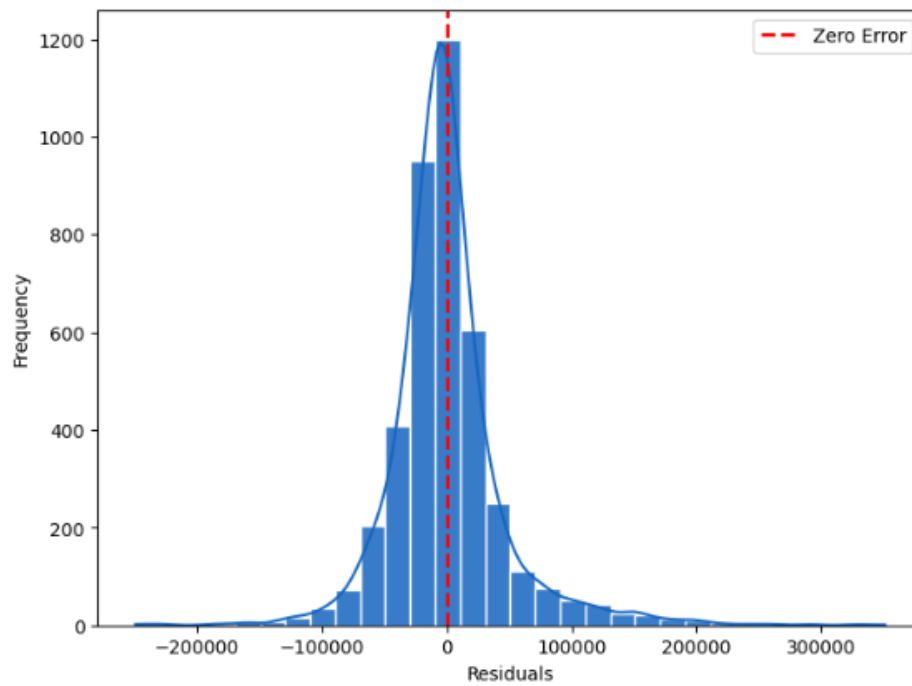


Figure 15: Tuned XG Boost Residual Distribution Graph

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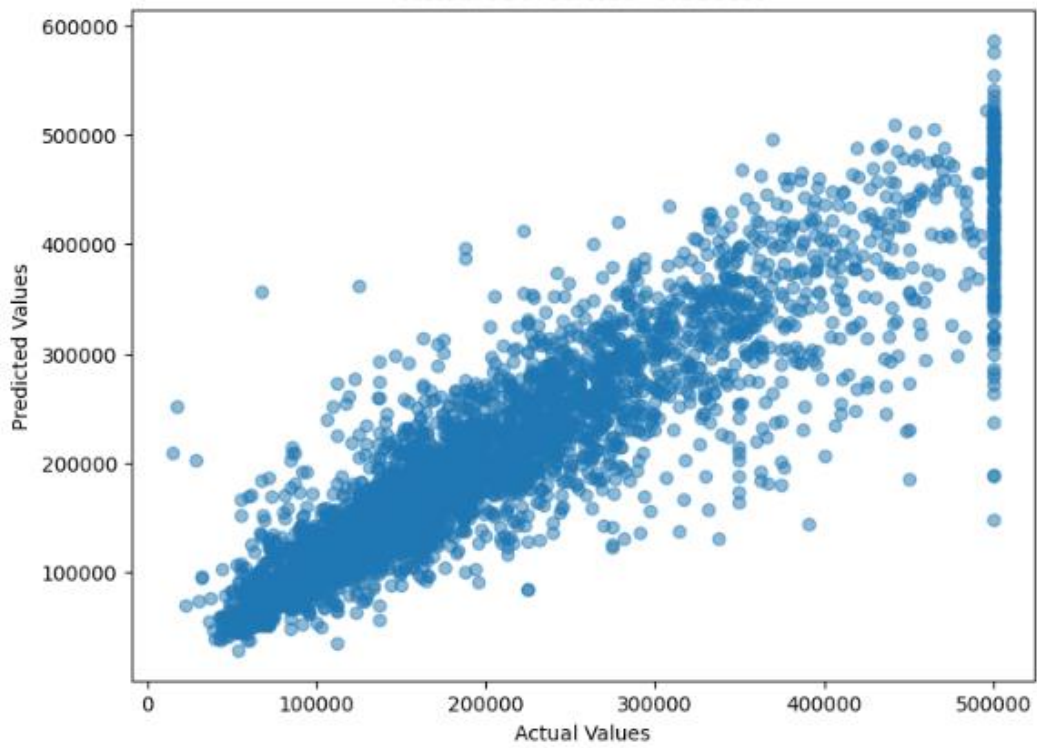


Figure 16: Tuned XG Boost Actual Vs Predicted Values Graph

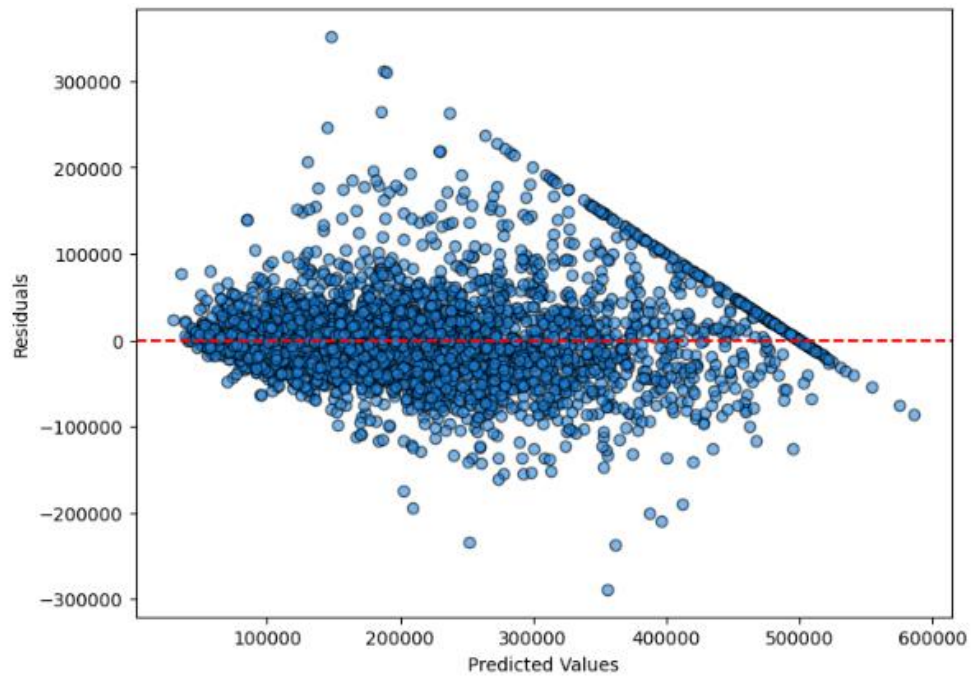


Figure 17: Tuned XG Boost Residual Scatter Plot Graph

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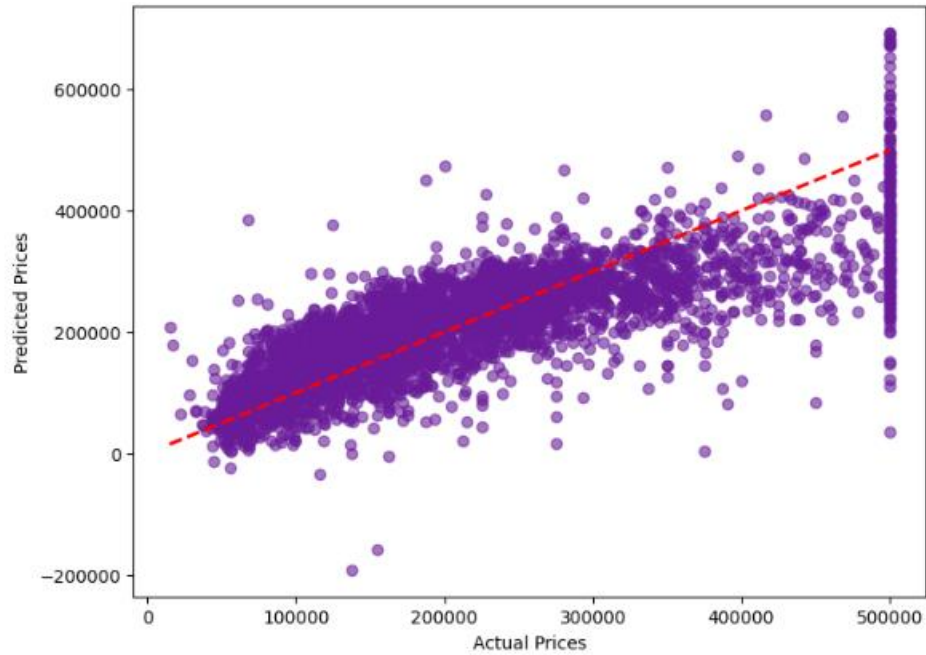


Figure 18: Tuned Ridge Regression Actual Vs Predicted Values

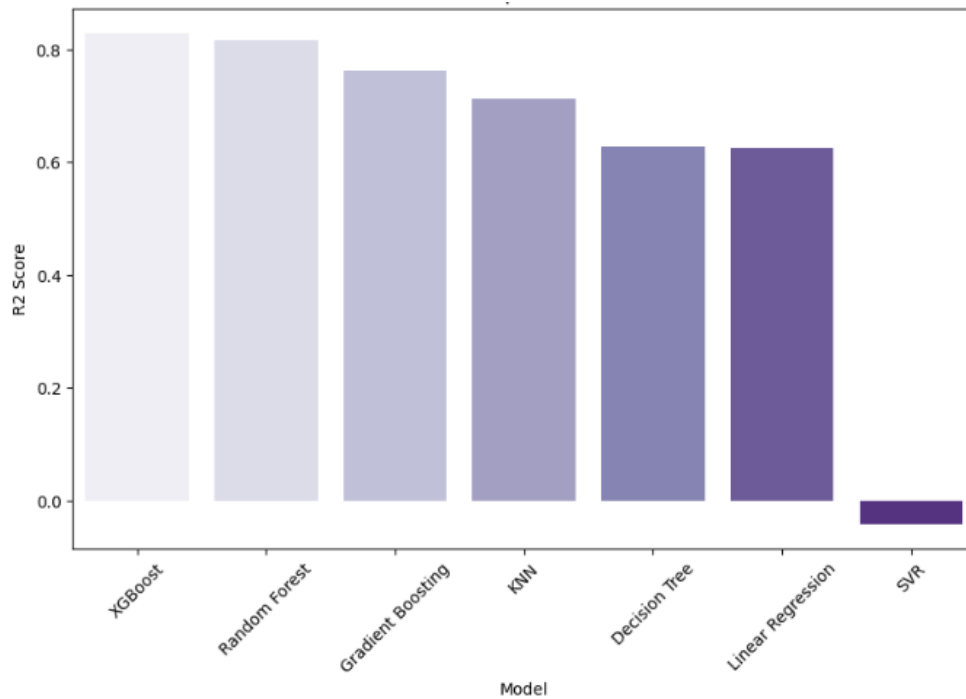


Figure 19: R² Score Comparison Graph between All Models

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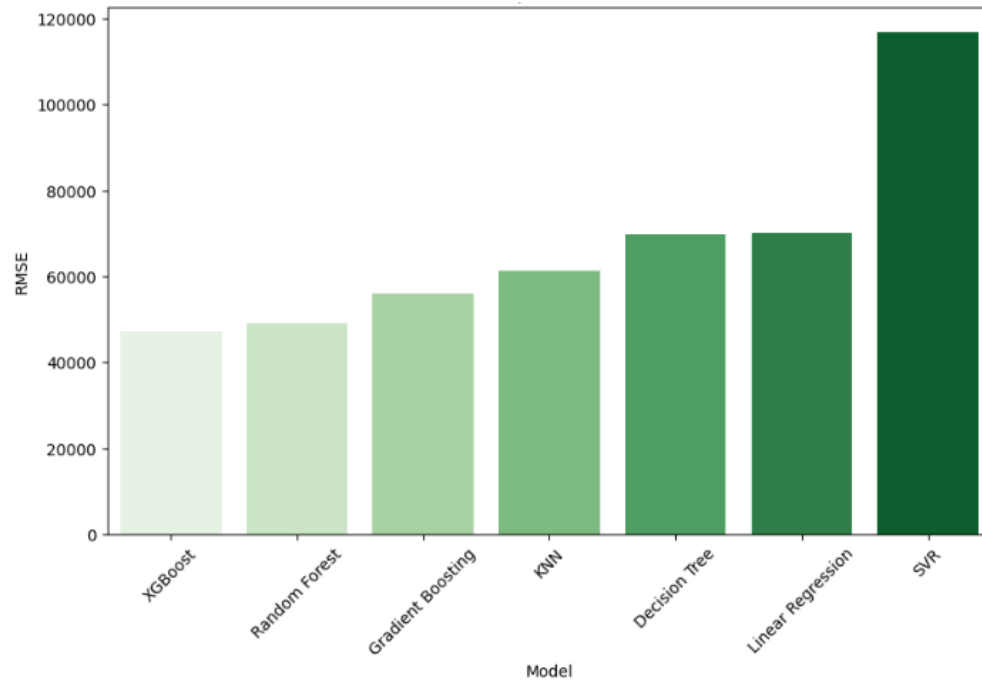


Figure 20: RMSE Comparison Graph between All Models

	Actual Price	Predicted Price
0	47700.0	53648.921875
1	45800.0	72783.460938
2	500001.0	474902.031250
3	218600.0	245135.546875
4	278000.0	273031.281250
5	158700.0	171483.875000
6	198200.0	274506.312500
7	157500.0	182561.156250
8	340000.0	280489.812500
9	446600.0	466959.562500
10	123200.0	140025.656250
11	253900.0	200074.984375
12	215100.0	179625.515625
13	220500.0	184397.734375
14	219800.0	238065.375000

Chart 8: Actual Vs Predicted Price

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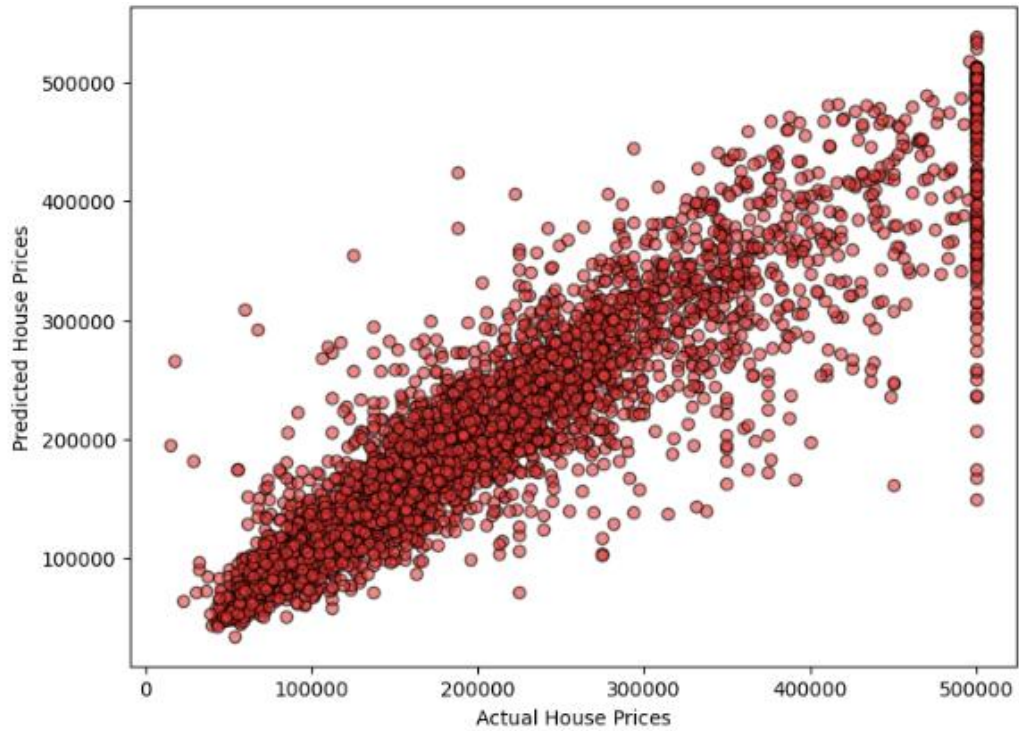


Figure 21: Tuned XG Boost Results of Predicting the House Prices

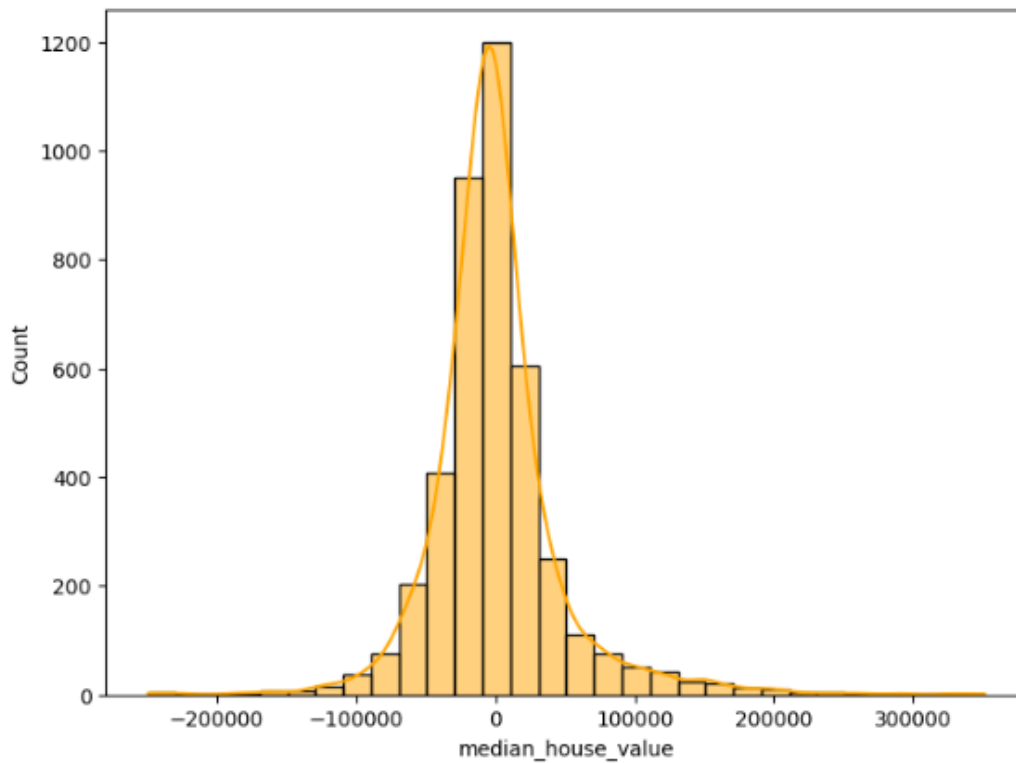


Figure 22: Prediction Error Distribution Graph, Tuned XG Boost

HOUSING PRICE PREDICTIONS

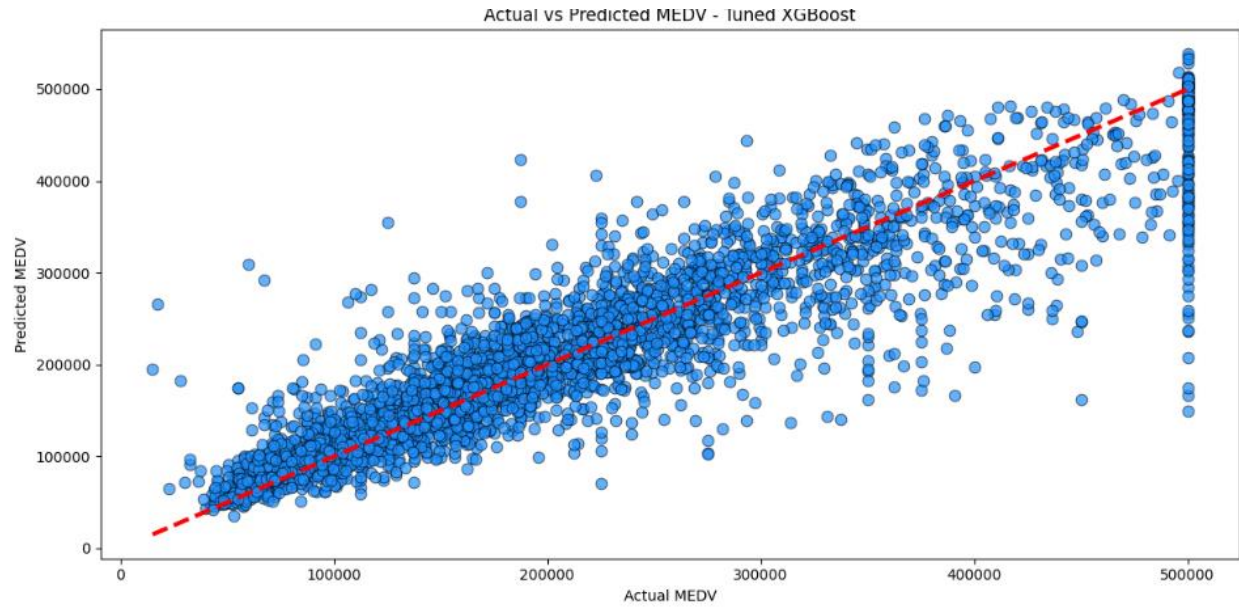


Figure 23: Tuned XG Boost, Actual Vs Predicted MEDV

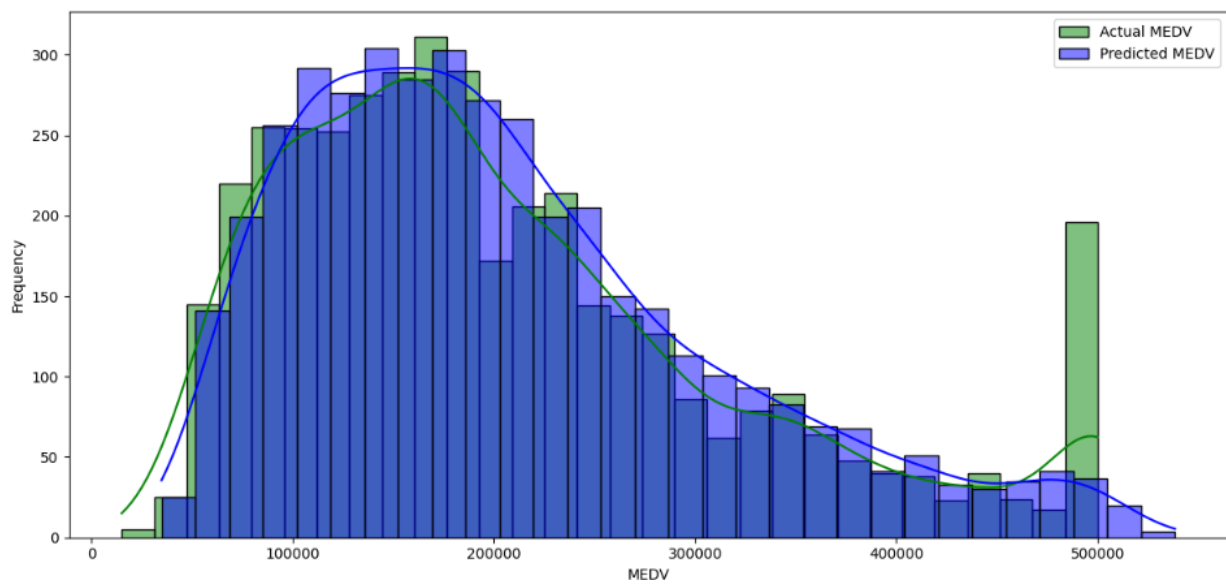


Figure 24: Tuned XG Boost, Distribution of Actual Vs Predicted MEDV

HOUSING PRICE PREDICTIONS

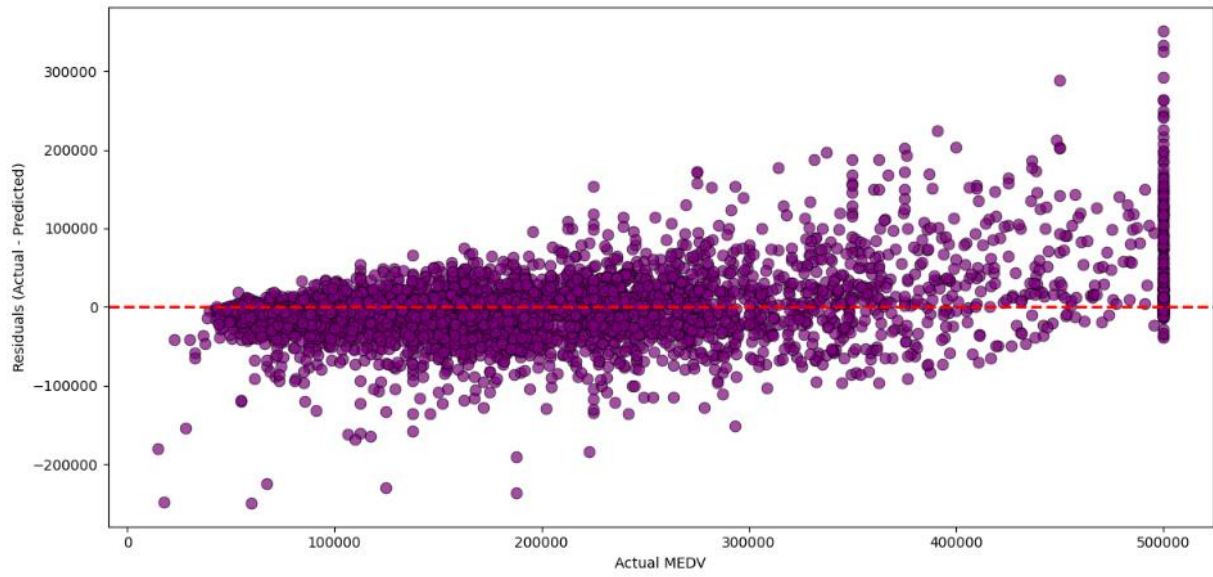


Figure 25: Tuned XG Boost, Residual Plots of Actual Vs Predicted MEDV

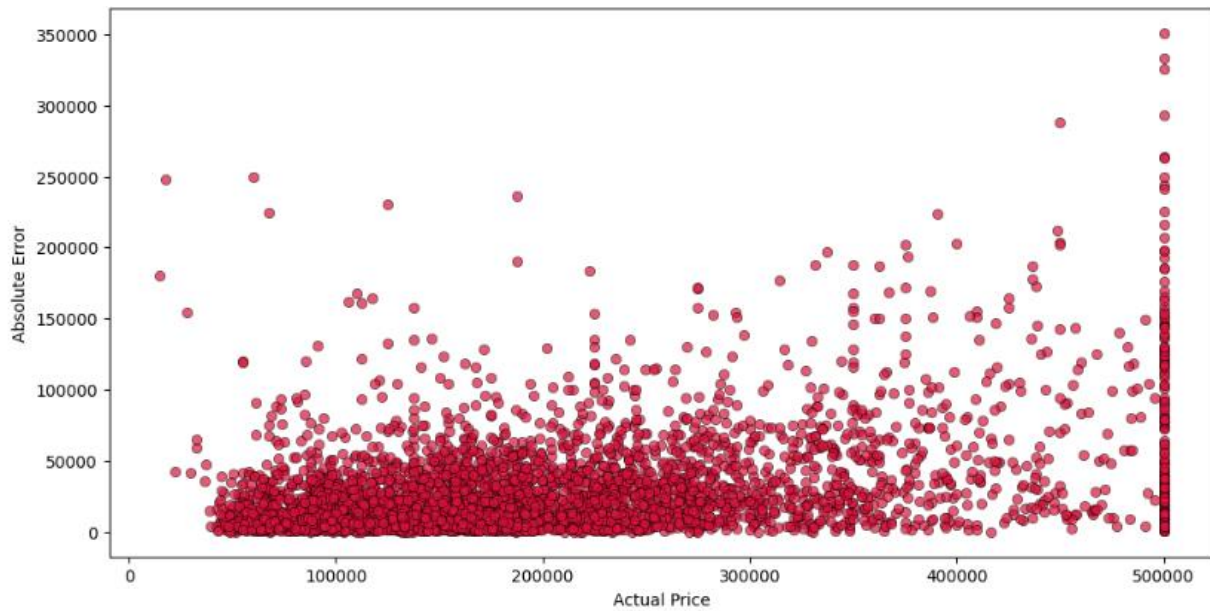


Figure 26: Tuned XG Boost, Error Magnitude Graph of Actual Vs Predicted Prices

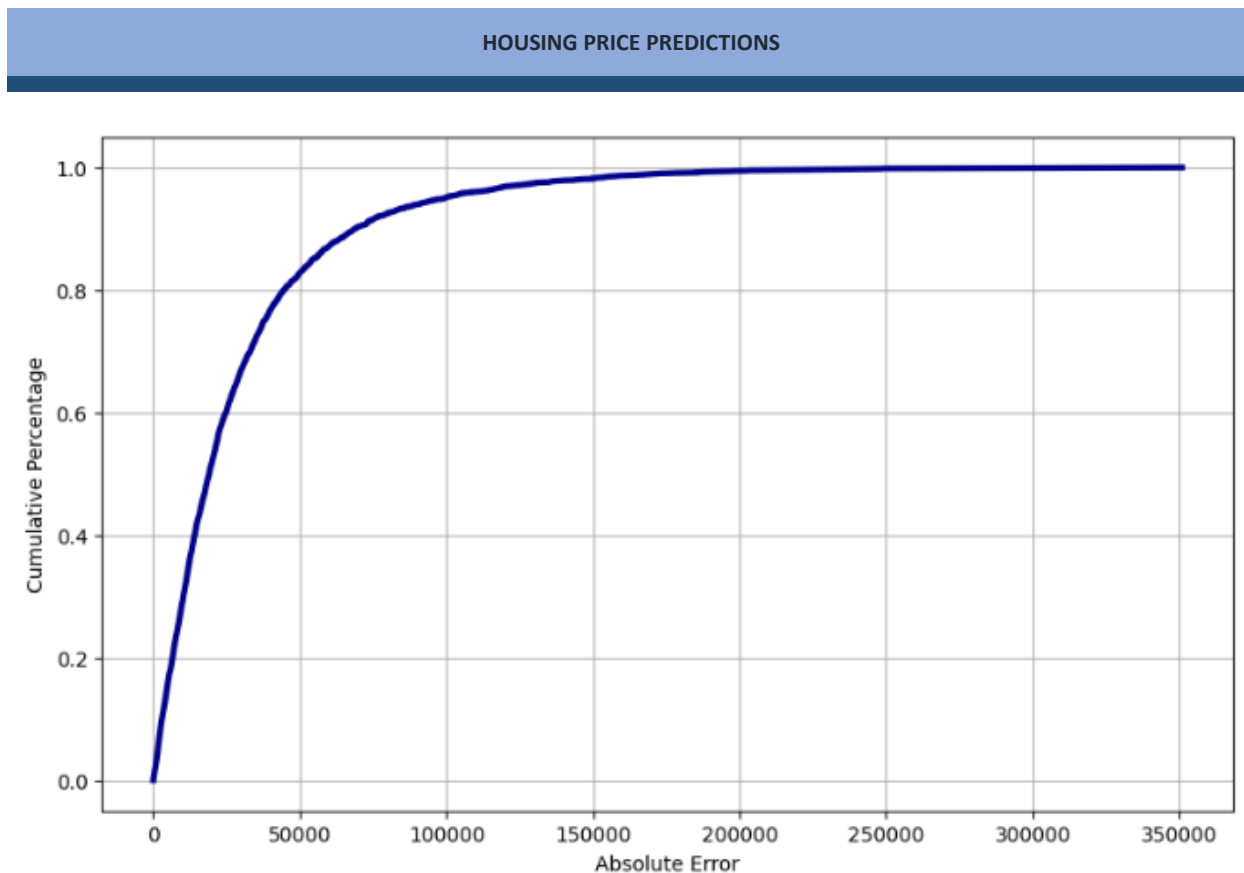


Figure 27: Tuned XG Boost, Cumulative Error Distribution Graph

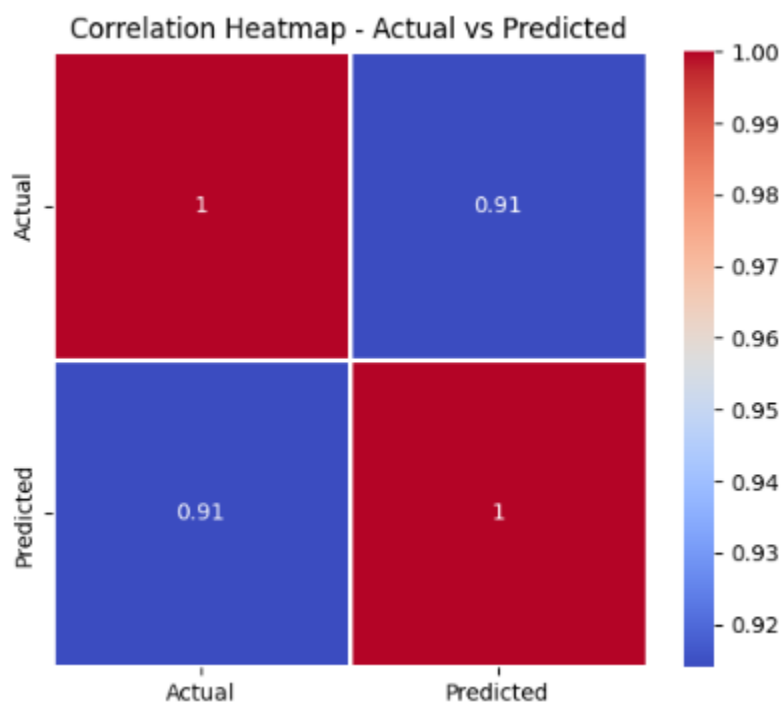


Figure 28: Tuned XG Boost, Correlation Heatmap of Actual Vs Predicted Prices

HOUSING PRICE PREDICTIONS

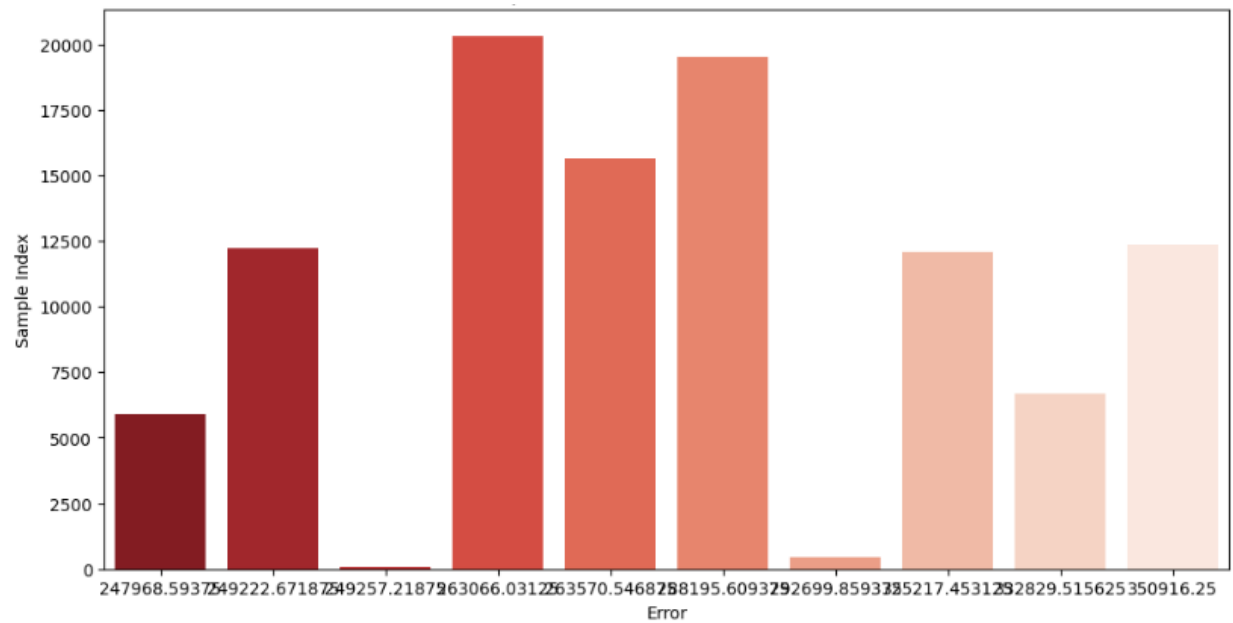


Figure 29: Top 10 Worst Predictions of Tuned XG Boost

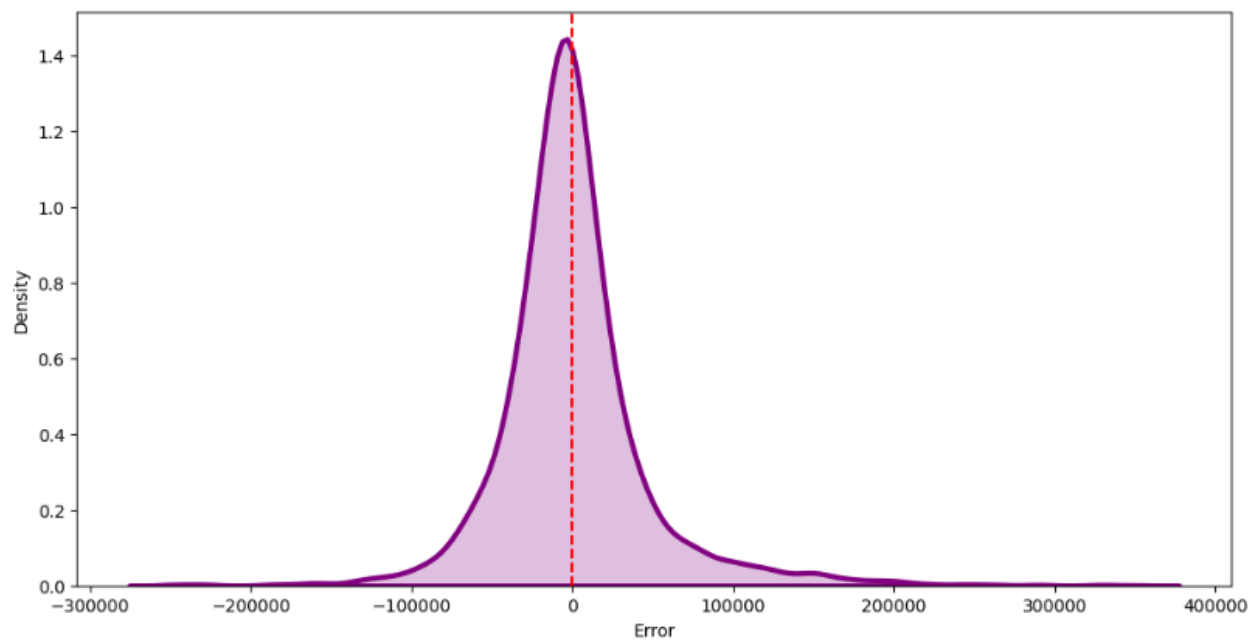


Figure 30: Residual Density Curve, Tuned XG Boost

HOUSING PRICE PREDICTIONS

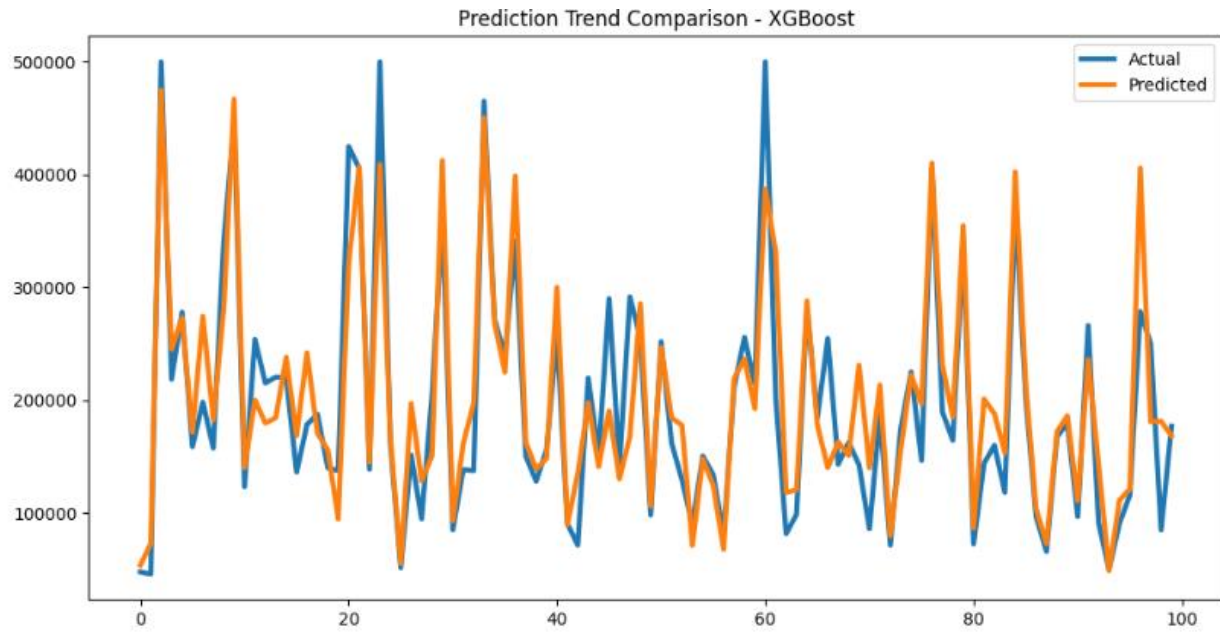


Figure 31: Prediction Trend Comparison, Tuned XG Boost

Conclusion

This project successfully demonstrates the application of machine learning techniques for housing price prediction. Multiple regression models were trained and evaluated, and the results clearly highlight the effectiveness of advanced ensemble methods in handling complex real-world data.

The key findings from the study are as follows:

1. **Median income** is the most important feature influencing housing prices, showing the strongest correlation with the target variable.
2. **Ensemble models** such as Random Forest, Gradient Boosting, and XGBoost perform better than traditional models due to their ability to capture nonlinear relationships.
3. **XGBoost** achieves the highest accuracy among all models, making it the most reliable predictor in this study.
4. **Hyperparameter tuning** significantly improves model performance by optimizing model parameters and reducing prediction error.

Overall, the system demonstrates strong predictive capability and good generalization performance, making it effective for real-world housing price estimation tasks.

Writer's Introduction



Assalam-u-Alaikum,

I am **Muhammad Anas Rathore**, a dedicated student currently pursuing my undergraduate studies in Computer Science. I maintain a strong academic record with a CGPA of **3.9** and continuously strive to expand my technical knowledge while applying innovative technologies to solve real-world challenges.

My primary areas of interest include **Artificial Intelligence, Machine Learning, Deep Learning, Data Science, and Software Development**. Throughout my academic journey, I have worked on multiple practical projects that have strengthened my technical expertise and enhanced my problem-solving abilities. These experiences have enabled me to bridge the gap between theoretical concepts and real-world applications.

In addition to my academic projects, I have gained valuable hands-on experience through internships and collaborative development work, where I contributed to software and AI-based solutions. These experiences have helped me strengthen my technical, analytical, and communication skills while working in professional environments.

As a lifelong learner, I remain committed to exploring emerging technologies, enhancing my skill set, and contributing to impactful innovations in the fields of Artificial Intelligence and Software Engineering.

A stylized, handwritten signature in black ink, appearing to read 'Anas Rathore'.

Muhammad Anas Rathore